

WORK COMPLETED BY
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Profiling Time Mismatch Resolution

Objective: To find which section of code is consuming more time apart from the QDFT time.

Approach used:

- Divided the entire code into sections and added timer to them.
- Removed the timer for those sections with least time contribution(time < 1 sec)
- Analyzed the profiling and did necessary subtraction of time so that it doesn't get counted double.

Anthracene(2e6o) Result

Profiling Time: 448.88 s

Real Time: 519.070 s

Gap: 70.19 s

Observations:

- Big gap between Profiled time and Real time in .err file.
- Other areas where time is spent need to be counted.

```
===== PROFILING SUMMARY =====  
DFT reference (SCF) time      : 45.55 s  
DFT energy evaluations time  : 9.80 s  
DFT Fock build time          : 9.26 s  
Hybrid embedding overhead    : 125.78 s  
Quantum VQE time              : 258.49 s  
-----  
Total measured time           : 448.88 s
```

```
real      8m39.070s  
user      29m2.970s  
sys       9m32.226s
```

New Profiling[Anthracene(2e6o)] Result

Profiling Time =
Library Import + Problem Setup & Integrals + Classical
Qiskit Overhead + QDFT Time

Major Time Spent:

1. Library Imports

2. Problem Setup & Integrals Time spent:

Transform Basis Sets
Construct the Quantum Problem
Initialize the Active Space
Calculate Initial Densities

3. Classical Qiskit Overhead:

Circuit Preparation
Transpilation & Compilation
Parameter Binding

Classical Qiskit Overhead = Solver Total - Quantum VQE

```
===== PROFILING SUMMARY =====  
Library Imports           : 38.17 s  
Driver Initialization     : 2.38 s  
Problem Setup & Integrals : 49.05 s  
-----  
Ansatz build              : 0.15 s  
Solver total (Pre/Post-processing time + Quantum VQE Time) : 233.39 s  
Classical Qiskit Overhead : 6.36 s  
-----  
DFT reference (SCF) time  : 45.34 s  
DFT energy evaluations time : 10.15 s  
DFT Fock build time       : 8.74 s  
Hybrid embedding overhead : 126.06 s  
Quantum VQE (callback only) : 227.04 s  
Total measured (QDFT only) : 417.33 s  
=====
```

```
real    8m36.372s  
user    28m35.536s  
sys      9m51.873s
```

Profiling Time: 510.41 s

Real Time: 516.372 s

Gap: 5.926 s

Pyrene(2e6o)

```
===== PROFILING SUMMARY =====  
DFT reference (SCF) time      : 78.87 s  
DFT energy evaluations time  : 16.17 s  
DFT Fock build time          : 19.18 s  
Hybrid embedding overhead    : 209.93 s  
Quantum VQE time             : 321.33 s  
-----  
Total measured time          : 645.47 s
```

```
real    11m43.899s  
user    45m47.821s  
sys     11m14.231s
```

Gap:58.429
s

```
===== PROFILING SUMMARY =====  
Library Imports      : 36.22 s  
Driver Initialization : 1.69 s  
Problem Setup & Integrals : 84.49 s  
-----  
Ansatz build          : 0.19 s  
Solver total (Pre/Post-processing time + Quantum VQE Time) : 246.95 s  
Classical Qiskit Overhead : 9.95 s  
-----  
DFT reference (SCF) time      : 78.78 s  
DFT energy evaluations time  : 17.43 s  
DFT Fock build time          : 18.69 s  
Hybrid embedding overhead    : 208.13 s  
Quantum VQE (callback only) : 237.00 s  
Total measured (QDFT only)   : 560.02 s  
=====
```

```
real    11m35.570s  
user    45m8.736s  
sys     11m17.601s
```

Gap: ~5 sec

Similarly tested for few other molecules(Benzene,propane,etc)

Secondary Work

- Obtained results of FCI,DFT and other methods(CASCI,CCSD,CASSCF,etc) for Anthracene(6e6o) using new ansatz.
- Also performed runs for Water molecule using 7 different exchange functionals and shifting max_itr to 500 to analyze with max_itr 50 results.
- Tabulated both obtained results.